TECHNIJIAN

CLIENT APPROVAL REPORT

OXPLive MCP Server

Secure AI Access to the OXPLive Platform

Prepared for	OrthoKinetix (ORX)
Prepared by	Technijian
Project	OXPLive Modernization — Phase B Deliverable
Date	June 2026
Status	Pending Client Approval — Phase F (Build)
Classification	Confidential — OrthoKinetix / Technijian Use Only

116	110	94	20	100%
Stored Procedures	API Endpoints	MCP Tools	Functional Domains	DB Call Coverage

01 Executive Summary

OrthoKinetix operates the OXPLive practice management platform — a mature SQL Server system running the full DME order lifecycle: intake, benefits verification, insurance authorization, shipping, billing, re-billing, and collections across 13 workflow queues.

Technijian proposes to make that data safely available to AI assistants (Claude) through a purpose-built Model Context Protocol (MCP) server. The server connects Claude directly to the OXPLive database via a set of 116 curated stored procedures across 20 functional domains — without granting the AI any direct table access.

The governing rule: every AI request follows a single, auditable path — Claude calls a documented MCP tool → the tool executes an approved stored procedure → the stored procedure reads or writes the database. No exceptions. No inline SQL. No table-level permissions for the AI service account.

This document presents the complete scope of what will be built, the 116-procedure coverage matrix, the security and HIPAA controls in place, the delivery plan, and the approval request to begin Phase F (Build). The full engineering specification — including every stored procedure, its parameters, return schema, API endpoint, and MCP tool definition — is provided in the companion STP-API-MCP Coverage Matrix.

02 Why Now — The Current-State Gap

Technijian has been the active IT and development partner for OrthoKinetix since approximately 2015. As part of the ongoing OXPLive Modernization project (migrating from ColdFusion to .NET 8 / React 18 under Technijian SDLC v6.0), a complete database-call inventory was conducted in early 2026.

What Was Found

The legacy OXPLive ColdFusion frontend makes hundreds of direct database calls across 13 workflow queues. These calls are not governed — they bypass stored procedures in many places, embed business logic in the UI layer, and contain hardcoded credentials. This creates three gaps that block safe AI access today:

- **No single governed access path.** The application reaches the database through a mix of stored procedures, inline SQL, and direct table reads. An AI assistant granted the same access would have uncontrolled write access to production data.
- **No audit trail by design.** Direct database calls leave no application-level record of who accessed what, when, or why — a HIPAA audit requirement.
- **No semantic boundary.** Without a procedure layer, there is no clean way to expose only what the AI should see while hiding raw PHI fields, financial data, and operational tables.

The Solution

The OXPLive Modernization addresses all three gaps by establishing a complete, SP-Only database access pattern. Every frontend interaction — and every AI tool call — must go through a named, documented, version-controlled stored procedure. This is both a modernization best practice and the prerequisite for safe AI integration.

03 Architecture — DB-Direct Design

The OXPLive MCP server uses a DB-Direct architecture: Claude connects to the MCP server via Streamable HTTP, and the server executes stored procedures directly against SQL Server via a secure connection pool. There is no intermediate REST API layer for MCP access.

Request Path

Claude (Desktop / Code / API)	*AI assistant — your staff or automated workflows*
Streamable HTTP POST /mcp	*MCP 2025-03-26 spec · session-keyed · optional API-key*
OXPLive MCP Server Node.js + mssql	*94 MCP tools · dispatches to named stored procedures*
SQL Server — oxadmin schema	*Service account EXECUTE-only on oxadmin schema · no table permissions*
OXPLive Tables · Views · Functions	*Production OrthoXpressDB — never directly accessible to the service account*

Why DB-Direct

The OXPLive Modernization Phase F will build a .NET 8 REST API, but the stored procedures are being designed now as part of Phase B. DB-Direct lets the MCP server go live the moment the stored procedures are tested — before the full API layer is complete — giving OrthoKinetix early AI access while the broader modernization continues.

Security posture is unchanged from an API-first design. The SQL Server service account (oxplive_mcp_svc) holds only EXECUTE on the oxadmin schema. It cannot read tables directly, run dynamic SQL, or access any object outside the curated procedure set.

Transport — Streamable HTTP

The server implements the MCP 2025-03-26 Streamable HTTP transport specification:

- **POST /mcp** — Creates or resumes a session; handles all JSON-RPC tool calls
- **GET /mcp** — SSE stream for server-initiated messages (session-keyed)
- **DELETE /mcp** — Gracefully closes a session
- **GET /health** — Returns { status: "ok", sessions: N }

04 Domain Coverage Matrix

The table below summarises the complete stored-procedure coverage across all 20 functional domains of the OXPLive system. Every domain corresponds to a distinct area of the OXPLive workflow UI. The full 116-row matrix — with individual SP parameters, return schemas, API endpoint definitions, and MCP tool specifications — is provided in the companion STP-API-MCP Coverage Matrix document.

#	Domain	STPs	API Endpoints	MCP Tools	Business Purpose
1	**Auth & Session**	5	4	2	User authentication, token lifecycle, session management
2	**Patient / Account**	18	16	16	Core patient record — demographics, insurance, job creation, medical history, status, cancellation
3	**Insurance Authorization**	5	5	5	Auth requests, line-item management, status tracking, approval workflow
4	**Queue & Workflow**	7	7	6	13-stage workflow queue (Intake → Billing → Collections); job transitions, queue counts
5	**Billing & Claims**	5	5	5	Fee schedules, billing search, claims processing, remittance management
6	**Scheduling**	10	9	9	Appointment creation, calendar search, therapist scheduling, 48-hour delivery alerts
7	**Patient Experience**	7	6	5	PSAT surveys, post-delivery call-back tracking, satisfaction scores, follow-up
8	**Collector Actions**	5	5	4	Daily collector call logs, AR action tracking, daily-total summaries
9	**Products & Inventory**	10	10	9	Equipment catalog, SKU management, inventory levels, assignment to patients at delivery
10	**Orders & Purchasing**	3	3	3	Purchase-order listing, vendor ordering, reorder tracking
11	**Documents / Files**	4	4	3	Document metadata, HIPAA-compliant signed download URLs for CMN, EDF, PHI files
12	**Directory, Users & Reps**	9	8	6	Doctor/facility directory, sales rep management, referring physician lookup, staff directory
13	**Time Records (HR)**	9	9	6	Timecard entry, manager approval workflow, payroll-ready export
14	**Lookups (Generic)**	4	4	2	Shared reference data — insurance payer names, territories, status codes, dropdown values
15	**Config**	3	3	2	System-wide configuration and company-settings retrieval
16	**Reporting**	4	4	4	Billing collections, PSR, variance analysis, KPI dashboard data
17	**Audit & Activity Logs**	4	4	3	HIPAA audit trail — user access log, data-change history, compliance events
18	**Compliance / Scheduled**	2	2	2	Mass job-move operations and compliance batch processing
19	**Yes/No Module (PA/CMN/EDF)**	1	1	1	Three compliance gates: PA status, CMN receipt, EDF sign-off — block queue progression
20	**Dashboard**	1	1	1	Real-time territory and user activity dashboard — session-aware, role-filtered
	TOTAL	**116**	**110**	**94**	**100% of OXPLive database calls covered**

STPs = Stored Procedures. API Endpoints = planned .NET 8 REST endpoints (Phase F). MCP Tools = tools exposed to Claude via DB-Direct MCP server. Some STPs are INTERNAL and are not exposed as MCP tools.

05 Security, HIPAA & Governance

OXPLive contains protected health information — patient demographics, insurance details, DME order data, and EDI 835/837 claims. Security controls are designed into every layer of this architecture, not added afterward.

Control	Implementation
Service Account Isolation	oxplive_mcp_svc holds only GRANT EXECUTE ON SCHEMA::oxadmin. No table-level SELECT, INSERT, UPDATE, or DELETE. No dynamic SQL is possible.
SP-Only Access Pattern	Every AI request resolves to a named, version-controlled stored procedure. Inline SQL from AI-generated queries is architecturally impossible.
Audit Logging	Domain 17 exposes audit-log procedures. HIPAA-sensitive SPs (Document_GetViewUrl, all patient-record writers) trigger server-side audit entries recording user, action, timestamp, and affected record ID.
PHI Field Masking	SSN and other PHI identifiers are masked at the stored-procedure level before they reach the MCP response payload. Patient-data fields require explicit role clearance.
Document URLs	File access (usp_Document_GetViewUrl) returns HIPAA-compliant signed URLs with short-lived expiry. No raw file paths or permanent links are returned.
Transport Security	Streamable HTTP runs over HTTPS in production. Optional MCP_API_KEY header provides HTTP-level bearer auth. Network access restricted by firewall to authorised IP ranges.
Session Isolation	Each Claude session receives a unique mcp-session-id. Sessions are server-memory-scoped; one session cannot access another session's state or results.
Compliance Domains	Domain 19 (Yes/No Module) enforces the three compliance gates — PA Status, CMN receipt, EDF sign-off — that must be satisfied before a job can progress. Domain 18 handles mass-move and scheduled compliance batch operations.

This design supports OrthoKinetix's HIPAA compliance posture. It does not replace the organisation's existing policies, Business Associate Agreements, or staff training, which remain in place.

06 What Gets Built

MCP Server — Node.js Application

The OXPLive MCP server is a Node.js / TypeScript application built on the official @modelcontextprotocol/sdk package. It uses the mssql library for SQL Server connectivity and Express for the Streamable HTTP transport.

- **DB Client —** Singleton mssql connection pool; executeSP(), querySP(), and querySPSingle() helpers that auto-prefix oxadmin. to every procedure name
- **20 Domain Tool Files —** One TypeScript file per domain, each exporting its MCP tool definitions and request handlers
- **Server Core —** createOxpliveMcpServer() registers all 94 tools and dispatches calls to the matching domain handler
- **Streamable HTTP Transport —** Express app with session map; supports multiple concurrent Claude sessions

SQL Server Service Account

A dedicated, least-privilege SQL Server login is created specifically for the MCP server. No further permissions are required. The service account cannot read, write, or alter any table, view, or function directly.

Provisioning script: CREATE LOGIN oxplive_mcp_svc WITH PASSWORD = '<vault-managed>'; CREATE USER oxplive_mcp_svc FOR LOGIN oxplive_mcp_svc; GRANT EXECUTE ON SCHEMA::oxadmin TO oxplive_mcp_svc;

Configuration & Deployment

All connection strings and secrets are managed via environment variables (never committed to source control). The server ships with a .env.example file and a config/claude-code.mcp.json snippet for immediate Claude Code integration.

What Is NOT in Scope

- The .NET 8 REST API layer (Phase F — planned, not yet approved)
- An admin portal or management dashboard
- Changes to the existing OXPLive ColdFusion application
- Changes to the OrthoXpressDB schema or existing stored procedures

07 Delivery Plan

The MCP server delivery is structured in three phases. Each phase ends with a clear, demonstrable result that OrthoKinetix can test before the next phase begins.

Phase	Deliverable	Scope	Exit Criteria
Phase 1	Stored Procedure Foundation	All 116 stored procedures authored, reviewed, and deployed to the OXPLive test database. Includes read-only procedures for all 20 domains plus write procedures for core entities.	All SPs pass unit tests against test DB; naming convention compliant; HIPAA-sensitive fields verified
Phase 2	MCP Server — Build & Test	Node.js MCP server with all 94 tools, DB-Direct wiring, Streamable HTTP transport, service account provisioning, and Claude Code integration.	Claude can invoke all 94 tools against test database; audit log confirms all calls recorded
Phase 3	Production Deployment	Production environment setup, credential vault integration, firewall rules, HTTPS, staff training session, and final sign-off from OrthoKinetix IT.	Server running in production; OrthoKinetix team can connect Claude and execute representative queries

Detailed timeline and effort estimates are confirmed during the kick-off session and provided in a companion Statement of Work upon approval of this document.

08 What We Need From You

Approval of this document authorises Technijian to begin Phase 1. A short kick-off session is scheduled immediately afterward to confirm the items below. None of these decisions block approval — they shape the detailed plan.

- **Domain priority order.** Which of the 20 domains should be built first? Suggested default: Domains 1–5 — Auth, Patient, Insurance Auth, Queue, Billing.
- **Test database access.** Confirmation that Technijian may continue working against the existing test/UAT environment for Phase 1 and Phase 2.
- **Service account provisioning.** Authorisation to create the oxplive_mcp_svc login and grant it EXECUTE on the oxadmin schema.
- **Credential vault.** Preferred secret-management approach (Azure Key Vault, Windows Credential Manager, or Technijian-managed vault) for the service account password and API key.
- **Hosting location.** Where the MCP server will run — on-premises IIS, Windows service, Docker container, or Technijian-hosted.
- **Internal users.** Who at OrthoKinetix will use Claude with the MCP tools, and at what initial permission level (read-only vs. read/write access to write-capable domains).

09 Approval & Sign-Off

By signing below, OrthoKinetix authorises Technijian to proceed with the OXPLive MCP Server project as described in this document, beginning with Phase 1 (Stored Procedure Foundation). This approval covers the technical approach, domain scope, and phased delivery plan.

Specific timeline, effort hours, and fees are confirmed in a companion Statement of Work, which will be issued within five (5) business days of receiving this signed approval.

OrthoKinetix	Technijian
Authorised on behalf of OrthoKinetix / OrthoXpress	Prepared and submitted by Technijian Inc.
_________________________ _____	Rohit Jain, Technijian June 2026
Name (print) Title	

Questions? Contact Technijian at 949.379.8499 (main) or rjain@technijian.com. The full engineering specification, STP-API-MCP Coverage Matrix, and Statement of Work are available on request.

A Appendix — Full Domain Reference

A.1 Domain Summary (authoritative counts)

#	Domain	STPs	API Endpoints	MCP Tools
1	Auth & Session	5	4	2
2	Patient / Account	18	16	16
3	Insurance Authorization	5	5	5
4	Queue & Workflow	7	7	6
5	Billing & Claims	5	5	5
6	Scheduling	10	9	9
7	Patient Experience	7	6	5
8	Collector Actions	5	5	4
9	Products & Inventory	10	10	9
10	Orders & Purchasing	3	3	3
11	Documents / Files	4	4	3
12	Directory, Users & Reps	9	8	6
13	Time Records (HR)	9	9	6
14	Lookups (Generic)	4	4	2
15	Config	3	3	2
16	Reporting	4	4	4
17	Audit & Activity Logs	4	4	3
18	Compliance / Scheduled	2	2	2
19	Yes/No Module (PA/CMN/EDF)	1	1	1
20	Dashboard	1	1	1
	TOTAL	**116**	**110**	**94**

A.2 The 13 Workflow Queues

The OXPLive platform routes DME orders through 13 named queues. Domain 4 (Queue & Workflow) provides the stored procedures that govern job transitions between queues.

#	Queue	Purpose
1	Intake	New DME orders received; initial eligibility check
2	Benefits	Insurance benefit verification in progress
3	Auth	Prior authorisation requested or under review
4	Pending	Awaiting patient or clinical information
5	Shipping	Order approved; equipment dispatch in progress
6	Billing	Delivered; claim submitted to payer
7	Re-Bill	Claim denied or adjusted; re-submission in progress
8	EDF	Equipment Delivery Form outstanding — compliance hold
9	Collections	Overdue balance; active collector follow-up
10	Yes/No	PA Status / CMN / EDF compliance gates pending
11	Inactive	Order on hold; no active work
12	Cancelled	Order cancelled; record retained for audit

13	Collector Actions	Collector daily-action tracking and totals

A.3 Companion Documents

- **STP-API-MCP Coverage Matrix —** All 116 stored procedures with full parameter lists, return schemas, API endpoint mappings, and MCP tool definitions. Provided as a separate A3-landscape PDF deliverable.
- **Statement of Work —** Issued within 5 business days of approval. Covers phased timeline, effort hours, and fees.